

## **Coding & Solution**

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|----------------------|------------------------------------------------------------|
| <b>Date</b>          | 01 JULY 2026                                               |
| <b>Team ID</b>       | XXXXX                                                      |
| <b>Project Name</b>  | Opticrop:Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                    |

### **Coding & Solution**

*This document presents the implementation details of the OptiCrop system. The solution demonstrates clean coding practices, modular implementation, and successful integration of Machine Learning with a Flask web application for crop recommendation.*

### **Solution Summary**

| <b>Field</b>                    | <b>Details</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>    | Github link : <a href="https://github.com/hasinibadviti/OptiCrop-Smart-Agricultural-Production-Optimization-Engine/tree/main">https://github.com/hasinibadviti/OptiCrop-Smart-Agricultural-Production-Optimization-Engine/tree/main</a><br>Live url : <a href="http://127.0.0.1:5000">http://127.0.0.1:5000</a>                                                                                                                          |
| <b>Programming Language(s)</b>  | PYTHON,HTML,CSS,JAVA SCRIPT                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Framework(s) Used</b>        | FLASK                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Key Features Implemented</b> | <ol style="list-style-type: none"> <li>1.Crop Recommendation Prediction using Machine Learning.</li> <li>2.Data preprocessing and feature scaling using Pandas and Scikit-learn.</li> <li>3.Interactive Flask web interface for user input and prediction.</li> <li>4.Real-time crop recommendation based on soil and environmental parameters.</li> <li>5.Model training, evaluation, and prediction using trained ML model.</li> </ol> |

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|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b><i>Pending / Incomplete Features</i></b> | <ol style="list-style-type: none"> <li>1. Weather API integration for live climate data.</li> <li>2. User authentication and prediction history.</li> <li>3. Mobile responsive UI and cloud deployment.</li> </ol>                                                                           |
| <b><i>Setup / Run Instructions</i></b>      | <ol style="list-style-type: none"> <li>1. Create a virtual environment.</li> <li>2. Install dependencies:<br/>pip install -r requirements.txt</li> <li>3. Run the Flask application:<br/>python app.py</li> <li>4. Open:<a href="http://127.0.0.1:5000">http://127.0.0.1:5000</a></li> </ol> |

**Code Quality Checklist**

| S.No | Criteria                                               | Status (Yes / No) |
|------|--------------------------------------------------------|-------------------|
| 1    | Code is modular and organized into functions / classes | YES               |
| 2    | Meaningful variable and function names are used        | YES               |
| 3    | Code includes comments / documentation where necessary | YES               |
| 4    | Error handling is implemented for critical operations  | YES               |
| 5    | The application runs without critical errors           | YES               |
| 6    | Code is committed to a version control repository      | YES               |

**Additional Notes / Comments:**

- ☐ Machine Learning pipeline includes preprocessing, model training, and prediction using Scikit-learn.
- ☐ Input validation and exception handling improve application reliability.